

NSCI0016 Literature Review

Elucidating the Correlation Between Bias-Induced Ion Migration and Charge Extraction Kinetics in Perovskite Devices

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Abstract

Metal-halide perovskites have been identified as very promising semiconductor materials for optoelectronic applications, especially for perovskite solar cells, owing to their high optical absorption and efficient charge transport. However, despite the rapid progress in power conversion efficiency, the device performance is often hindered by bias-driven ion migration, which causes current–voltage hysteresis, transient instabilities, and degradation. This review aims to summarize the current status of research on the correlation between bias-driven ion migration and charge extraction kinetics in perovskite devices. By integrating experimental and theoretical research, this review demonstrates that ion migration dynamically modulates the internal electric fields and interfacial energetics, resulting in history-dependent charge extraction and recombination dynamics. The results underscore the crucial role of considering perovskite devices as ionic–electronic hybrid systems and offer insights for material and device design approaches to enable kinetics-limited performance.

1 Introduction

As a newly emerged semiconductor material, metal-halide perovskites have produced highly promising performance results in optoelectronic devices such as perovskite solar cells (PSCs) due to their outstanding optical absorption and long carrier diffusion lengths. Although power conversion efficiency (PCE) has been improved rapidly in the past few years, approaching those of established photovoltaic technologies, the long-term operational stability of perovskite devices remains a challenge. Among the mechanisms limiting device performance and stability, bias-induced ion migration and charge extraction kinetics have attracted increasing attention, yet their correlation is not fully understood.

Bias-induced ion migration is an electric-field-driven motion of halide vacancies or mobile ions within the perovskite lattice under an applied bias during device operation. This process leads to a redistribution of ionic charge, particularly near interfaces between the perovskite absorber and charge transport layers. Such redistribution can lead to internal electric fields and band bending. On the other hand, charge extraction kinetics describes the rate and efficiency with which photogenerated charge carriers are transferred from the perovskite absorber to the electron and hole transport layers before recombination occurs. Efficient charge extraction is essential for achieving high short-circuit current density, fill factor, and overall device efficiency. When ion migration alters interfacial energetics or defect states, it can directly influence charge extraction pathways, leading to time-dependent device behaviour such as current–voltage hysteresis and performance degradation.

The primary aim of this research topic is to elucidate the correlation between bias-induced ion migration and charge extraction kinetics in perovskite devices through bias-dependent steady-state and transient optoelectronic characterisation. Specifically, this work aims to quantify how ionic redistribution under applied bias alters charge extraction rates, interfacial recombination dynamics, and transient photocurrent and

photovoltage responses, and to investigate to what extent do ion-mediated effects contribute to efficiency losses and operational instability in perovskite solar cells.

Understanding this correlation has significant technological and socio-economic implications. The global perovskite solar cell market was valued at roughly USD 264.6 million in 2024 and is projected to expand to over USD 7 billion by 2030, implying a compound annual growth rate (CAGR) exceeding 70% in the next few years.[1] Other forecasts similarly indicate robust growth with estimates placing the market at several billion dollars by 2034 with a strong CAGR between 30% and 40%.[2] This dramatic expansion reflects increasing investment, commercial interest, and anticipated deployment of perovskite-based modules in utility, residential, and building-integrated applications. Improving efficiency, lowering manufacturing costs, and advancing operational stability through fundamental studies are key to realising these socio-economic benefits.

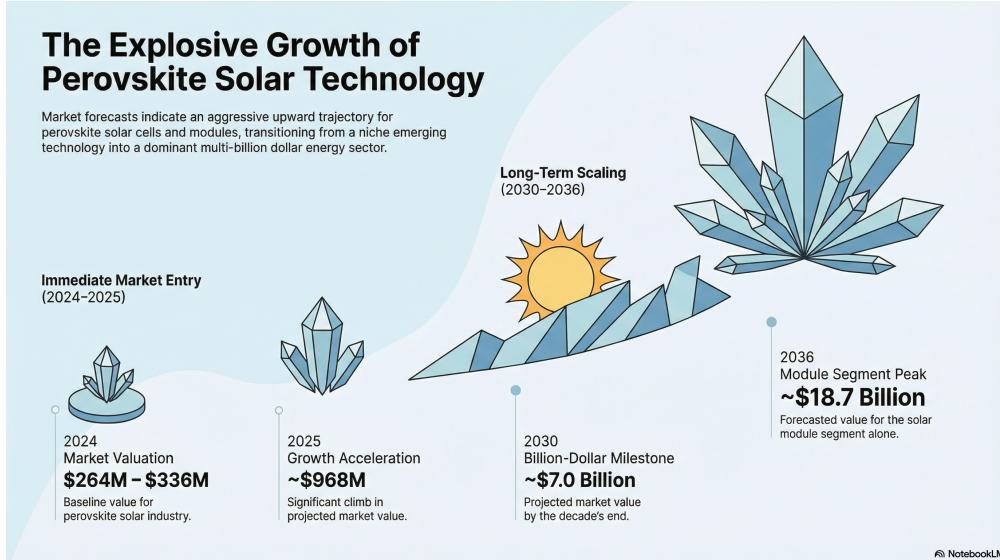


Figure 1: Perovskite market growth prediction data from 2024 to 2036 [1][2][3], plot generated using NotebookLM.

In addition, perovskites' solution processability at low temperatures allows deposition on flexible and lightweight substrates, making them well-suited for flexible electronics such as wearable sensors and bendable photodetectors. Furthermore, the presence of heavy elements such as lead and iodine gives perovskites strong X-ray absorption coefficients, while their long carrier diffusion lengths and high charge-collection efficiency enable high-sensitivity X-ray detectors operating at low bias and potentially lower cost than conventional semiconductor detectors. Together, these attributes make perovskite as a versatile material for multifunctional devices beyond traditional solar energy conversion.[3]

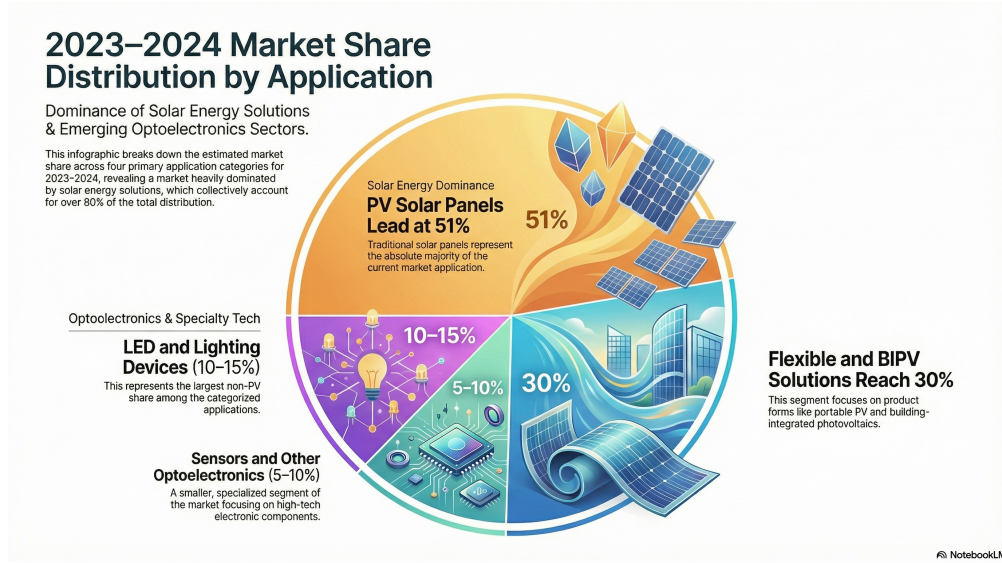
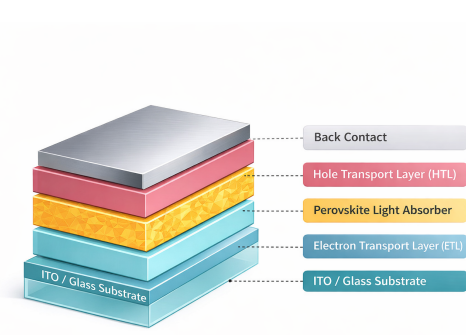


Figure 2: Perovskite market share distribution in 2023-2024 from [4][1], plot generated using NotebookLM.

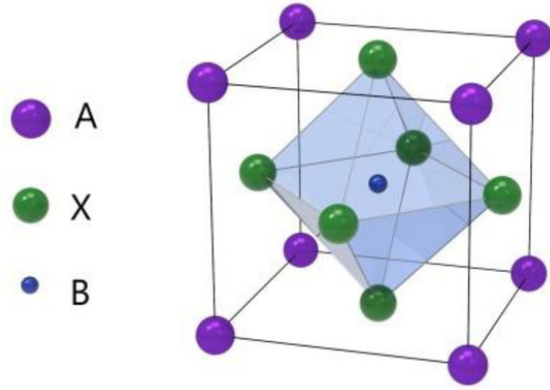
2 Review

2.1 Metal-Halide Perovskite Materials and Device Physics

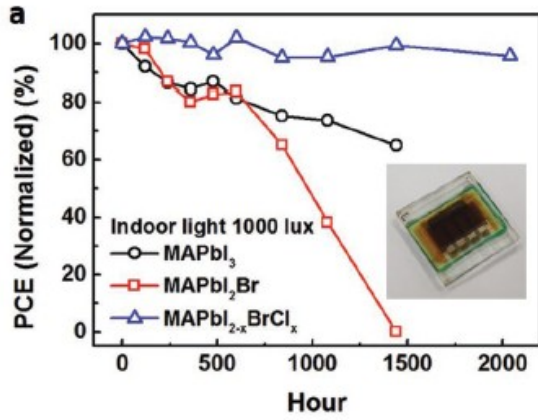
Metal-halide perovskites have the general structure ABX_3 , where A is a non-metal cation, B is a metal cation, and X is a halide anion (e.g. F^- , Cl^- , Br^- , I^-). This crystalline structure endows perovskites with strong light absorption and excellent charge transport properties. Within a few years, PSC has achieved PCE from around 12% to over 25%. By 2020, single-junction PSC reached over 25% efficiency, comparable to established solar technologies.[5] In terms of device architectures, perovskite devices can be fabricated in planar heterojunction or mesoporous structures, meaning perovskite infiltrating a porous TiO_2 or Al_2O_3 scaffold. A typical device stack includes a transparent conducting oxide, electron transport layer (ETL), perovskite absorber, hole transport layer (HTL), and metal electrode.



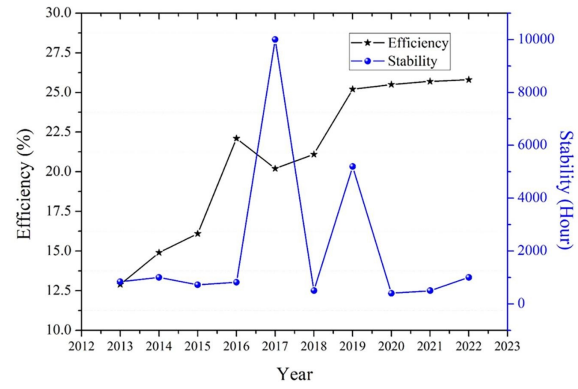
(a) PSC structure breakdown. Figure generated using NotebookLM.



(b) ABX_3 perovskite crystal structure.[6]



(c) Time dependent PCE. [7]



(d) Evolution of PSC efficiency and stability from 2012-2022 [5].

Figure 3: Representative figures illustrating perovskite solar cell structure, perovskite crystal structure, and efficiency evolution.

Under illumination, perovskites generate free electron-hole pairs due to their low exciton binding energy and high dielectric constant. To extract charges, the opposite charged carriers will separate and drift through the perovskite layer to the charge-selective contacts via internal electric field and diffusion. However, this requires the carrier diffusion length exceed the absorber thickness. In other words, the carriers need to be able to travel long enough to the charge-collecting layer. Efficient charge extraction requires careful energy level alignments at interfaces between each layer. Nevertheless, the polycrystalline films can sustain micron-scale carrier diffusion lengths, allowing efficient charge collection despite the presence of grain boundaries, which resist charge carrier movement.[8]

Material	PV Context	Typical Diffusion Length	Measurement Unit
Crystalline silicon (c-Si)	c-Si solar cells	100 -- 300	μm
Perovskite	single-crystal MAPbI_3	> 175 (1 sun); > 3 (weak light)	μm ; mm
Perovskite	polycrystalline MAPbI_3 films	~ 1	μm
GaAs	n-type range	0.65 -- 3	μm
CdTe	thin film PV	0.75 -- 1.5	μm
CIGS ($\text{Cu}(\text{In,Ga})\text{Se}_2$)	thin film	0.5 -- 1.25	μm
Amorphous silicon (a-Si:H)	intrinsic device quality	0.1 -- 0.3	μm
Organic semiconductors (OPV)	Exciton diffusion length	5 -- 20	nm

Figure 4: Data collected from [8][9][10][11][12][13][14][15][16]. Table plotted using NotebookLM.

2.2 Ion-Migration-Induced J–V Hysteresis

One of the most intriguing aspects of perovskite devices is the anomalous hysteresis observed in current–voltage (J–V) scans. When scanning the voltage in opposite directions, the resulting curves do not retrace, indicating slow internal processes. Researchers quickly linked this to mobile ionic species within the perovskite that move in response to electric fields. Under an applied bias, negatively charged ions (e.g. I^- or Br^-) drift toward the positive electrode, while positive ions (e.g. MA^+ or vacancies acting as effective positive charge) migrate toward the negative electrode. This ionic redistribution occurs on timescales from milliseconds to seconds, lagging behind the fast electronic response and thus yielding hysteretic J–V behavior.

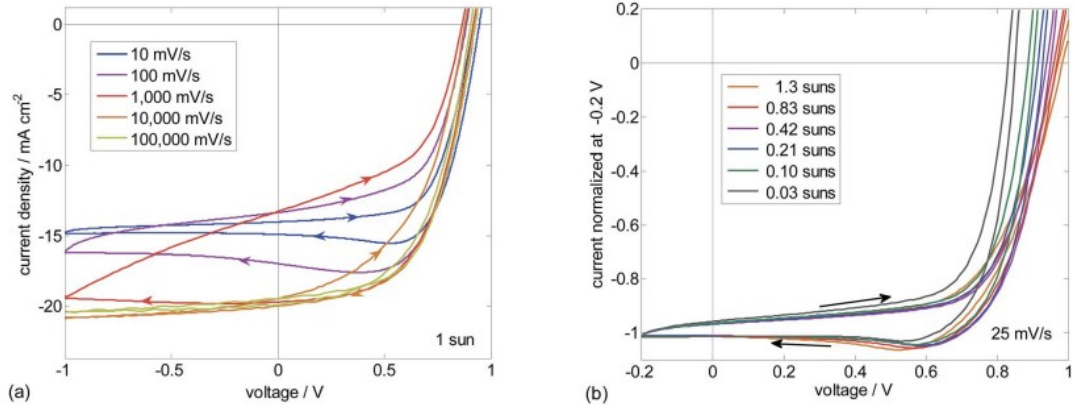


Figure 5: Plot obtained by Tress et al. (2015) [17] Current–voltage curves of TiO_2 based $CH_3NH_3PbI_3$ devices with pronounced hysteresis. Arrows denote the sweep direction.

Some early work hypothesized ferroelectric polarization as the cause of hysteresis [18][19], but the weight of evidence now points to ion migration as the dominant mechanism [20][21]. Uncertainties remain in identifying the exact migrating species and their diffusion coefficients. Iodide ions or iodide vacancies are widely considered the primary movers, but cation migration and even movement of defects like interstitials can also occur [18][22][23]. Moreover, the role of grain boundaries as fast ion diffusion pathways is an ongoing topic. Studies show that single-crystal perovskites with no grain boundaries exhibit minimal hysteresis, implying grain interfaces are a key factor for ionic motion. [7]

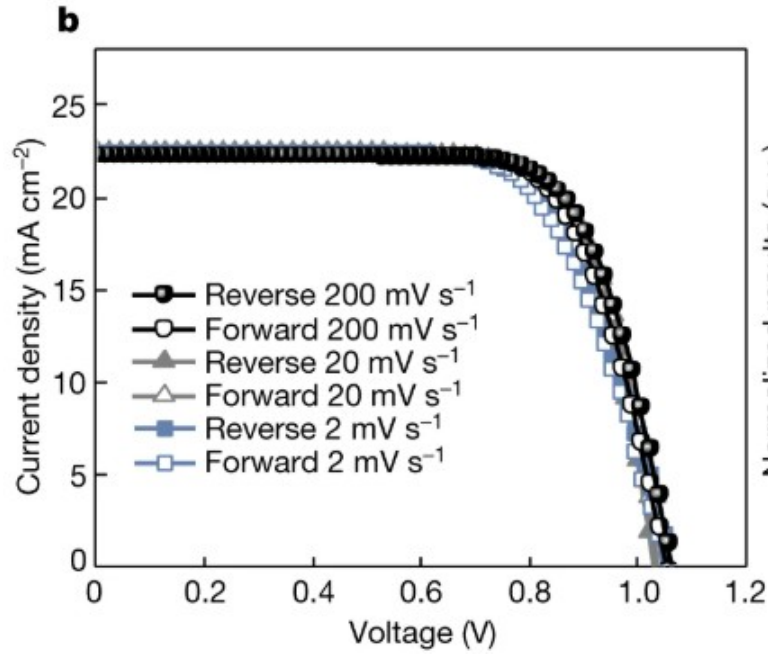


Figure 6: Plot obtained from Lei et al. (2020) [7] Thin $MAPb_{0.5+x}Sn_{0.5-x}I_3$ single-crystal devices maintain stable J–V curves even at very low scan rates.

2.3 Bias-Dependent Charge Extraction Kinetics

When an external bias is applied, mobile ionic species (primarily halide ions or vacancies) redistribute within the perovskite absorber and accumulate at charge-selective interfaces. As a result, the internal electric field experienced by photo-generated carriers becomes history dependent.[24] For example, forward-bias conditioning near the open-circuit voltage can drive negatively charged ions toward the ETL interface, partially screening the built-in field. Upon switching the device to short-circuit conditions, this ionic space charge remain transiently present, therefore reducing the effective driving force for electron extraction and leading to a delayed increase in photocurrent until ionic redistribution is restored. Alternatively, reverse-bias preconditioning can cause a transient increase in the internal electric field during short-circuit operation, resulting in an initial photovoltage overshoot that decays as ionic redistribution proceeds.[25] These phenomena clearly illustrate that carrier extraction in perovskite solar cells is not exclusively an electronic process but is instead intimately linked to the slow dynamics of ionic diffusion.

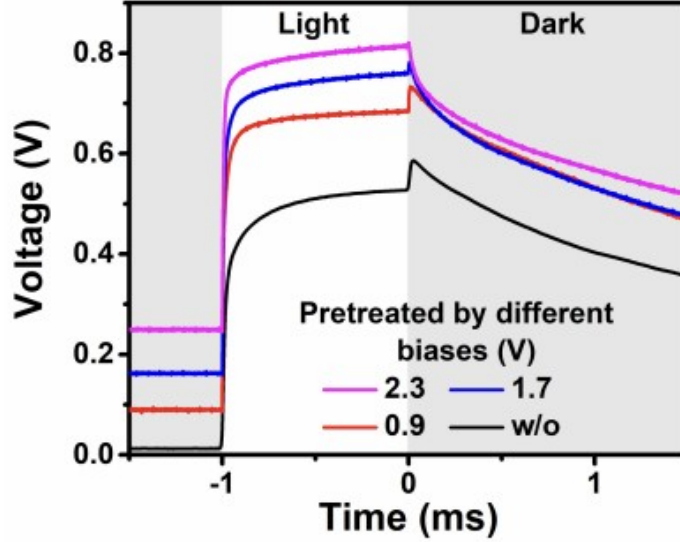


Figure 7: Transient photovoltage (TPV) measurement of PSC devices obtained by Li et al. (2018) [25] Demonstrating the overshoot effect.

2.4 Time-Resolved Probes of Ionic–Electronic Coupling

Time-resolved electrical measurements provide direct experimental evidence for the ion migration and charge extraction coupling. Transient photocurrent and photovoltage studies have shown that perovskite solar cells display a delayed relaxation towards a steady-state current or voltage after sudden bias voltage changes, which is a consequence of the finite ion migration times. Unger et al. [26] first reported this significant current and voltage transients after bias reversal and suggested that this could not be explained by purely electronic RC effects but was instead attributed to mobile ions responding to the altered electric field. Complementary impedance spectroscopy measurements have provided evidence of a large low-frequency capacitance in perovskite solar cells, which

is typically several orders of magnitude larger than the geometric capacitance. This phenomenon is known as the “giant dielectric response” and has been attributed to the accumulation of ions at interfaces, which effectively acts as an ionic capacitor in parallel to the electronic junction. As a result, sudden changes in the load or bias voltage result in current or voltage tails and overshoots due to the charging or discharging of the ionic capacitor. These findings clearly demonstrate that ion migration introduces additional kinetic barriers to charge extraction, making the device response strongly dependent on bias voltage history and timescales, rather than purely electronic transport properties.

2.5 Ion-Mediated Trade-Offs Between Extraction and Recombination

The ion redistribution under bias not only affects the extraction field but also influences the balance between charge extraction and recombination. Ionic space-charge regions can create internal electric fields that either assist or oppose carrier drift toward the electrodes, hence altering carrier residence times within the absorber.[17] If ionic accumulation produces an unfavorable field configuration, photogenerated carriers may stay longer in the perovskite bulk or near interfaces, increasing the probability of non-radiative recombination and reducing collection efficiency. This has been realized during open-circuit and forward-bias operation, where devices show increased hysteresis and degraded performance until ionic relaxation occurs.[18] However, ionic redistribution is not always deleterious. In some cases, mobile ions can partially passivate charged defects or grain-boundary traps through local charge compensation, thus lowering recombination losses and improving the photovoltage.[27] The overall effect of ion migration on charge extraction kinetics thus represents a competition between beneficial screening effects, which can lower recombination, and deleterious field effects, which can hinder carrier transport. Crucially, this competition is a function of both the timescale of measurement and the device operating point, thus emphasizing the dynamic nature of ion-electron interactions in perovskite solar cells.

2.6 Conceptual Implications

Cumulatively, these results make it clear that bias-driven ion migration is a key process in determining the kinetics of charge extraction in perovskite devices. Ionic concentration changes lead to time-dependent changes in internal electric fields, interfacial energies, and the presence of slow kinetic rates that coexist with fast electronic transport. Consequently, the efficiency of extraction, the rate of recombination, and device performance become bias-dependent and timescale-dependent. This understanding reinterprets hysteresis phenomena in perovskite devices not as a curiosity but as a symptom of a more fundamental ion–electron coupling that controls device kinetics. From a materials perspective, approaches that inhibit ion migration or use ion migration control such as compositional engineering, defect passivation, and the use of single-crystal or quasi-single-crystal absorbers can be used to stabilize charge extraction kinetics and achieve kinetics-limited performance. However, a complete understanding of perovskite devices also necessitates the inclusion of ionic degrees of freedom in models of electronic transport.

2.7 Open Questions and Future Directions

Despite great progress in the understanding of bias-driven ion migration in perovskite solar cells, several basic questions still remain unanswered, especially concerning the optimal balance between ionic and electronic dynamics for high performance and stability. The main challenge is to distinguish between the slow ionic dynamics and the fast electronic transport without completely suppressing ion migration. While ion migration is commonly linked to negative effects like hysteresis and transient instability, it may also have positive effects, such as the compensation of charged defects or self-healing processes in the perovskite lattice. This raises the fascinating possibility that a controlled level of ion migration could be tolerated or even harnessed as long as it does not affect charge extraction. Whether there is an optimal ion migration rate that reduces defect-mediated recombination while maintaining electronic performance remains an open question. Adding to the complexity is the incomplete knowledge of the mobile ions involved. While halide ions or vacancies are commonly regarded as the primary mobile ions, cation migration, especially of methylammonium, can become important at high temperatures or under illumination, where chemical decomposition and volatilization may occur. The observation of metal cation migration under strong bias conditions, as well as the migration of extrinsic ions from transport layers, adds to the complexity. Highly advanced characterization tools, including time-of-flight secondary ion mass spectrometry and operando X-ray-based analysis, are increasingly being used to clarify the identity of the mobile ions and their timescales. In terms of materials and device engineering, a variety of methods have been proposed to suppress ion migration, including compositional engineering with mixed-halide or all-inorganic perovskites, grain boundary passivation to suppress fast diffusion channels, and the use of two-dimensional or quasi-two-dimensional capping layers to physically confine ions. However, these methods commonly introduce trade-offs, where modifications that improve ionic stability may deteriorate electronic performance. Ion immobilization without compromising electronic performance is therefore a key materials challenge. Furthermore, most experimental investigations of ion-mediated extraction and recombination dynamics are performed under static or preconditioned conditions, whereas actual device operation entails dynamic bias variations resulting from intermittent illumination or shading. A deeper understanding of the role of ionic redistribution under dynamic device operation conditions on charge extraction efficiency and degradation timescales of hours to days is critical to unlocking laboratory stability timescales for actual device lifetimes. Notably, recent experiments on single-crystal perovskite devices have shown that near-zero ion migration rates can lead to remarkably stable device performance under continuous illumination, although similar stability under scalable polycrystalline device architectures remains highly challenging. Ultimately, the challenge of achieving actual device lifetimes satisfying the decades-long operational requirements for commercialization will require not only better ion migration control but also advances in encapsulation, thermal management, and device architectures. A more enabling framework that combines theoretical modeling of ionic-electronic transport with cutting-edge characterization will be essential to enable the rational design of perovskite materials and devices that either suppress or harness ion migration for superior functionality.

To systematically study the ion migration and charge extraction kinetics coupling, an

experimental program can be constructed around the use of Fluxim PAIOS (Photo-Analysis I–V, Transient, and Operational Spectroscopy) analysis tools. PAIOS is a platform that combines the capabilities of steady-state and transient optoelectronic analysis. With PAIOS, one can carry out bias-dependent current–voltage (J–V) analysis with well-defined preconditioning bias conditions (such as forward, reverse, and open-circuit biasing conditions), followed by transient photocurrent (TPC) and transient photovoltage (TPV) analysis to determine characteristic timescales for rise and decay under various ionic conditions. Additionally, PAIOS allows impedance spectroscopy analysis, which can be used to determine the magnitude of low-frequency ionic contributions to device capacitance, potentially separating electronic and ionic contributions to device response. By designing an experimental program that systematically varies bias preconditioning, light intensity, and temperature, it is possible to construct real-time maps of how ionic redistribution modifies internal fields and charge extraction processes.

In addition to the experimental design, a structured modelling part can be formulated to interpret and forecast device response to dynamic bias. Numerical simulations using coupled ionic drift–diffusion and electronic transport models can be set up in tailored Python code, driven by the experimental parameter space identified through PAIOS analysis. This two-fold experimental and modelling approach not only offers a quantitative understanding of the coupling between ion migration and extraction kinetics but also offers a set of data from which benchmarking parameters (such as ionic relaxation time constants, effective extraction rates, and recombination parameters) can be derived and compared across material platforms or device architectures. These experimental and analysis steps can be organized sequentially, starting with initial steady-state analysis, followed by dynamic transient analysis, and finally model refinement, offering a clear roadmap for future research.

3 Conclusion

This review has examined the current status of the literature on bias-driven ion migration and its role in charge extraction kinetics in perovskite devices. Taken together, the existing literature clearly shows that ion migration is a fundamental material property of metal-halide perovskites due to their ionic lattice structures, and that ion migration is a key factor in determining current–voltage hysteresis, transient, and stability properties. The experimental results obtained from steady-state measurements, time-resolved photocurrent and photovoltage measurements, and impedance spectroscopy measurements have clearly confirmed that ion migration can dynamically alter the internal electric fields and interfacial energy levels, resulting in history-dependent charge extraction and recombination properties. Recent work on single-crystal perovskite devices has further confirmed that ion migration can be suppressed to achieve stable and scan-rate-independent device performance, supporting the notion that many non-ideal device properties are of ionic origin rather than purely electronic origin.

Nevertheless, there are still important areas of uncertainty in the present understanding of ion–electron coupling in perovskite devices. Although the role of halide ions or vacancies is presently believed to be the most important one, the role of cation migra-

tion, metal ion motion, and extrinsic ions from transport layers is still not clarified. Moreover, although most of the work is focused on phenomenological observations of hysteresis or transient effects, there is still a lack of quantitative correlations between ionic redistribution, charge extraction rates, and recombination kinetics. The absence of models that explicitly describe ion drift–diffusion and electronic transport makes it difficult to gain predictive understanding of device performance.

The literature reviewed highlights several strengths of the field, including the use of advanced time-resolved and operando characterization techniques to probe ionic effects, as well as successful materials and interface engineering strategies that mitigate ion migration. However, these strengths are accompanied by inherent limitations. Many mitigation approaches such as compositional tuning or the introduction of two-dimensional capping layers involve trade-offs between improved ionic stability and reduced electronic transport. Moreover, a significant portion of experimental work is conducted under static or preconditioned conditions, which may not accurately reflect real device operation involving dynamic bias changes and fluctuating illumination.

One of the major challenges that has arisen from the literature is the requirement to balance ionic stability with efficient charge extraction. This is not only a matter of improving materials design and interfaces, but also of developing a time-dependent theoretical model that captures the dynamics of ionic and electronic processes. This will be critical in ensuring that laboratory-scale performance and stability are translated into perovskite devices that are capable of meeting the long-term operational requirements necessary for commercialization.

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